Big Data Analysis and Visualization of Global Superstore Sales using Cloud Tools

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Aim :

To perform comprehensive analysis and visualization of global superstore sales data in order to identify key trends, patterns, and insights related to sales performance, customer behavior, and regional distribution.

Objectives :

- To extract and load dataset for analysis
- To preprocess and clean the data
- To perform exploratory data analysis
- To identify sales trends and patterns
- To create interactive visualizations and dashboards

Tools and technologies used :

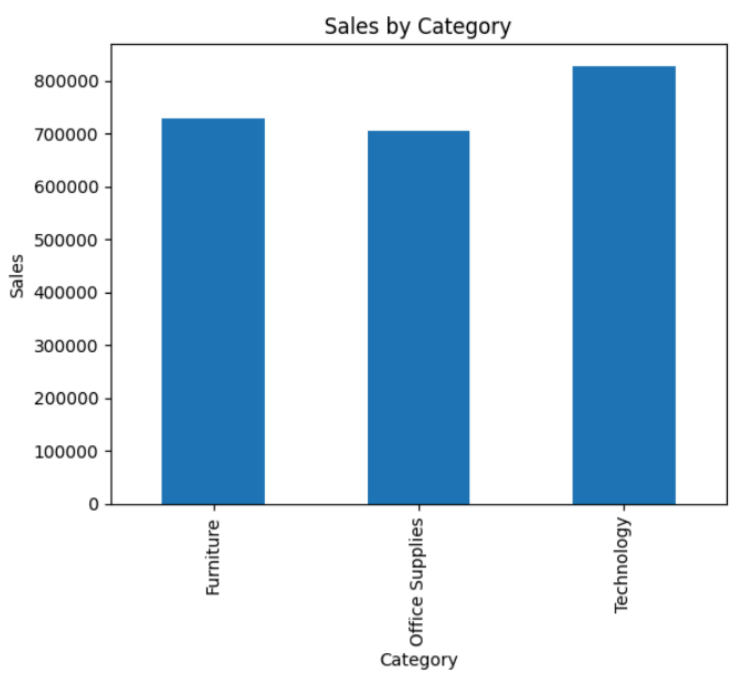
Dataset : Global superstore data
Platform : Google Colab
Cloud Service Provider : Google Cloud
Data Engine : Python (Pandas, NumPy)
Visualization tool : Matplotlib
BI tool : Power BI

Step-by-Step Implementation :

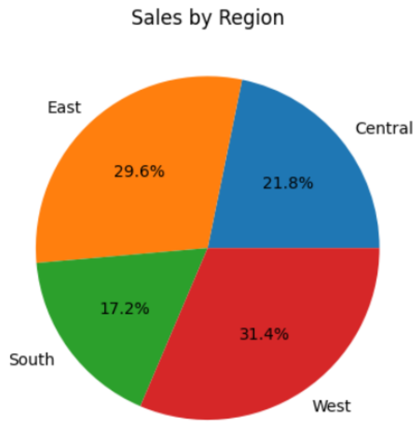
1. The dataset was collected and uploaded into the working environment.
2. Data was loaded using Python libraries such as Pandas.
3. Data preprocessing was performed including handling missing values and formatting columns.
4. Date column was converted into proper format for time-based analysis.
5. Exploratory Data Analysis (EDA) was conducted using groupby operations.
6. Various insights were extracted such as sales by category, region, and product.
7. Visualization charts were created using Matplotlib.
8. The dataset was then imported into Power BI.
9. Interactive dashboard was created including slicers and multiple visual charts.

Data Analysis

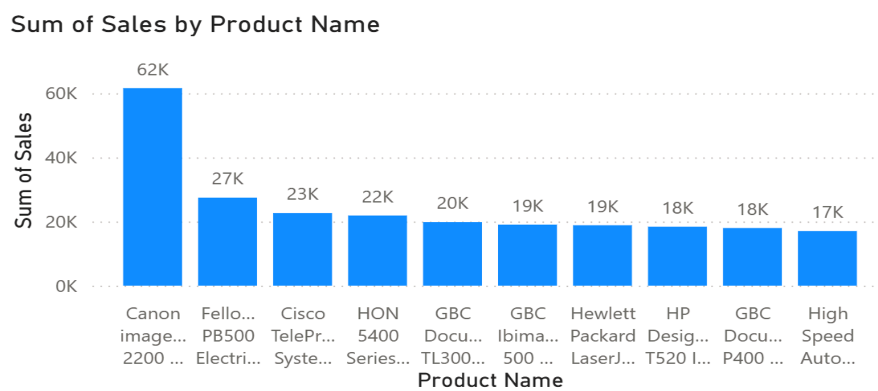
- **Sales by Category:**
  Technology category shows the highest sales contribution.



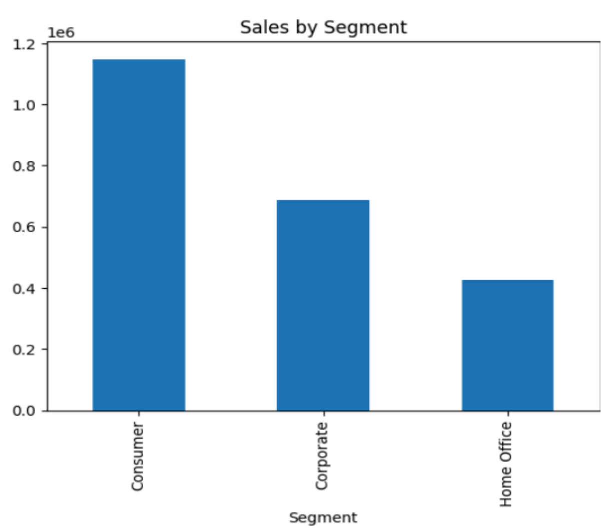
- **Sales by Region:**
  Western region contributes significantly more compared to others.



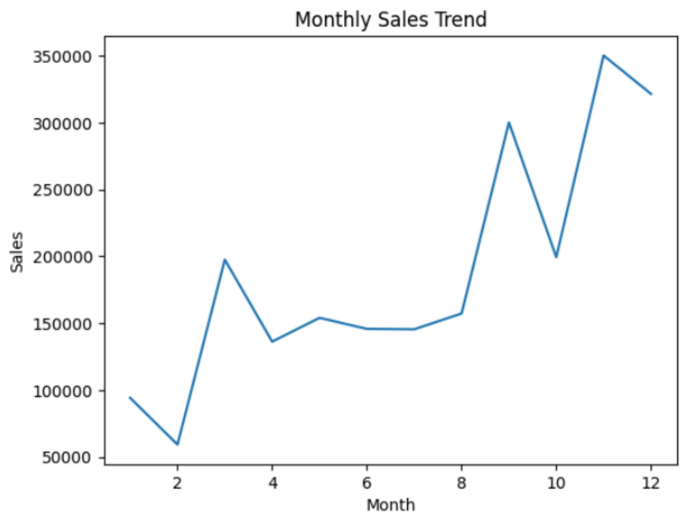
- **Top Products:**
  A small number of products generate a large portion of total sales.



- **Sales by Segment:**
  Consumer segment dominates overall sales.



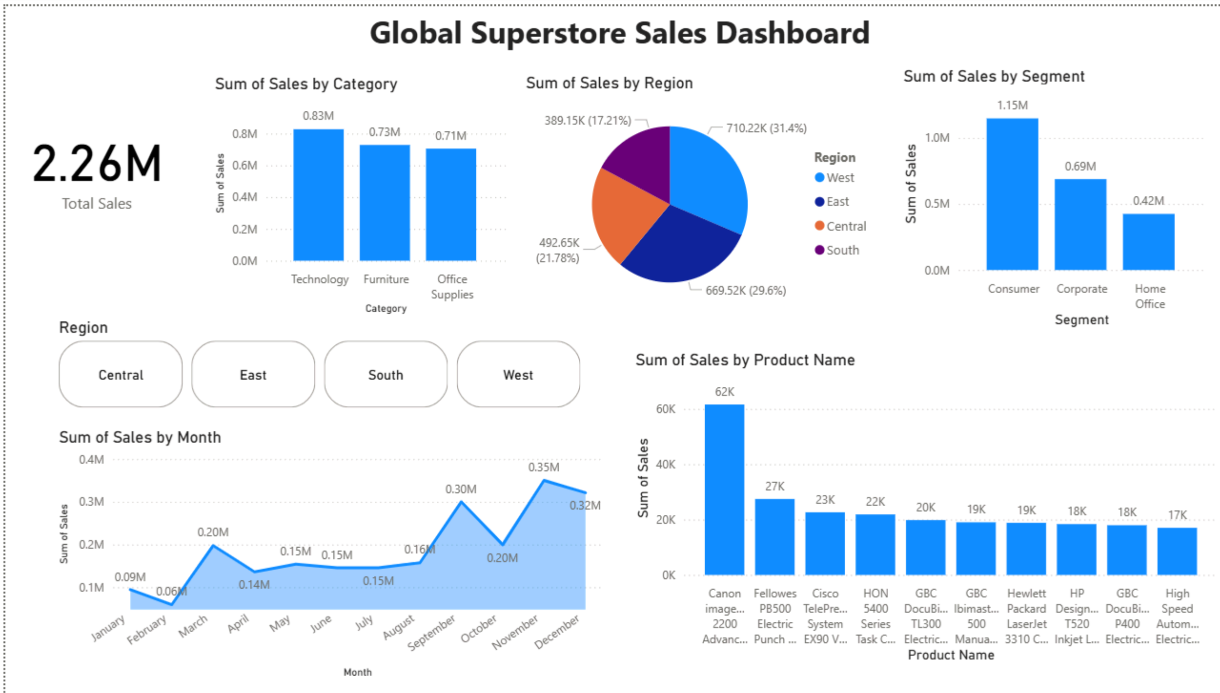
- **Monthly Sales Trend:**
  Sales show variation across different months indicating seasonal trends.



Data Visualization Reports

The following visualizations were created:

- Bar chart showing sales by category
- Pie chart representing regional sales distribution
- Line chart showing monthly sales trend
- Bar chart displaying top 10 products by sales
- Dashboard with slicers for interactive filtering



These visualizations help in understanding patterns and making comparisons effectively.

Outcomes :

- Successfully analyzed the global superstore dataset
- Identified important sales trends and patterns
- Built meaningful visualizations for better understanding
- Created an interactive dashboard for data exploration
- Gained practical experience in data analysis and visualization tools

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Conclusion :

The project demonstrates how data analysis and visualization techniques can be used to extract meaningful insights from large datasets. The use of dashboards enhances decision-making by presenting data in a clear and interactive manner.